

LUIE: Learnable physical model-guided underwater image enhancement with bi-directional unsupervised domain adaptation

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ABSTRACT

Recently, learning-based underwater enhancement (UIE) methods have made considerable progress, significantly benefiting downstream tasks such as underwater semantic segmentation and underwater depth estimation. Most existing unsupervised UIE methods utilize the atmospheric image formation model to decompose underwater images into background color, transmission map, and scene radiance. However, they rely on simplified physical models for estimating the transmission map, over-simplifying its complex formation, which results in imprecise modeling of underwater scattering effects. Additionally, supervised UIE methods heavily depend on synthetic data or ground truth, leading to limited generalization capabilities due to the substantial domain gap presented in different underwater scenarios. To tackle these challenges, we propose a Learnable physical model-guided unsupervised domain adaptation framework for Underwater Image Enhancement, dubbed LUIE. LUIE learns to predict background light, depth, and scene radiance from an underwater image. We incorporate a learnable network to estimate the transmission map based on the predicted depth map. To minimize the inter-domain gap between synthetic and real underwater images, we introduce a bi-directional domain adaptation method that alternates the background light from each domain. Experimental results demonstrate the effectiveness of our proposed method compared to existing approaches, and high-level experiment results validate that our enhanced underwater results. Experiments in real-world settings on underwater ROVs platform with NVIDIA Jetson AGX Xavier further confirm the effectiveness and efficiency of our work.

1. Introduction

The quality of underwater images often suffers from numerous problems, such as water scattering, absorption, non-uniform illumination with color casts, low contrast, and complex degradation, which create challenges for downstream tasks, including underwater semantic segmentation [1–3], underwater SLAM [4–7] and object detection [8–11]. Moreover, these challenges also impact the ability of underwater Remotely Operated Vehicles (ROVs) to explore and gather information about the underwater environment, including tracking marine species migration [12], monitoring coral reefs [13], and inspecting submarine cables and wreckage [14], among others. *Therefore, Efficient, robust, and accurate underwater image enhancement techniques are crucial for the downstream task of underwater scenarios and inspection of underwater ROVs.*

In recent years, significant progress has been made in learning-based methods for underwater image enhancement. While Supervised

methods, utilizing deep neural networks (DNNs), have shown promising results [16–19]. However, the complexity of underwater image degradation types and subjectivity in image labeling have led to the exploration of fusion methods [15,20]. Despite their successes, supervised methods still have limitations in terms of generalization and computational efficiency. To address these challenges, semi-supervised [21] and unsupervised methods [22,23] have been proposed. *Semi-supervised methods leverage both real-world and synthetic images, while unsupervised methods make use of prior knowledge or transfer learning from synthetic to real underwater domains.*

The normally used prior knowledge—simplified atmospheric scattering model [24] can be used to mathematically describe underwater scenes as follows:

$$I(x) = J(x)t(x) + A(1 - t(x)), \quad (1)$$

Here, $I(x)$ represents the underwater image, $J(x)$ denotes the scene radiance, A represents background light, and $t(x)$ is the transmission

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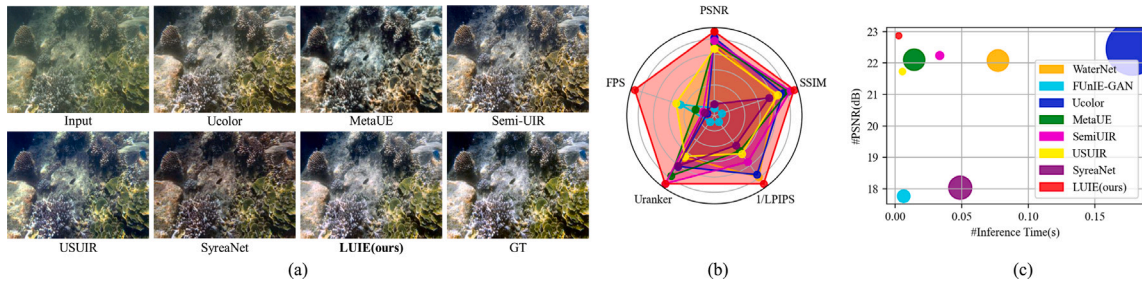


Fig. 1. (a) presents the qualitative comparisons of SoTA methods and our method on UIEB dataset [15]. (b) provides the numerical scores for five types of measurement metrics, including enhancement evaluation metrics (PSNR, SSIM, LPIPS), underwater image quality assessment metric (Uranker), and computational efficiency metric (FPS), tested at a resolution of $1,024 \times 1,024$. (c) illustrates the correlation between model parameters (represented by the size of circles), inference time, and PSNR in comparison to different methods.

map given by $e^{-\beta d(x)}$, where β and $d(x)$ represent the attenuation coefficients and depth map, respectively. The objective of UIE is to estimate $J(x)$ from $I(x)$. Some synthetic datasets [25] are available that are generated using this model, and some UIE methods use it to decompose underwater images [19,22]. However, these UIE methods simplify the construction of the transmission map due to the uncertainty of β , which results in a simplified representation of the true mapping of the depth map as the transmission map. This uncertainty often results in inaccurate intermediate predictions, leading to error accumulation and affecting the final results. Moreover, the design of decomposition networks typically involves a high computational cost, making it challenging to directly use them on underwater ROVs.

In this work, we propose a novel learnable physical model-guided unsupervised underwater image enhancement approach with bi-directional domain adaptation named LUIE, as illustrated in Fig. 2. We create a compact network guided by a physical model. Specifically, we decompose the underwater image into background light, scene radiance, and depth map using three individual sub-networks (named B-Net, J-Net, and D-Net, respectively). Since the attenuation coefficient β is genuinely unknown, we parameterize the mapping from depth to transmission map with another sub-network (T-Net). The model is trained with a bi-directional intra- and inter-domain adaptation method. To maintain intra-domain consistency, we reconstruct the underwater image using background light, scene radiance, and transmission map according to Eq. (1) and supervise the network with LPIPS loss between the reconstructed and original underwater images. We also shuffle the background light of the synthetic and real domains to minimize the domain gap between the synthetic and real domains. It is worth noting that each of our sub-networks is extremely compact, consisting of only five layers of CNN, and only J-Net is utilized for radiance recovery during inference, resulting in a significantly reduced computational cost and inference time.

Unlike existing datasets that rely on simplified indoor models or GANs [20], we developed the Underwater3k dataset using Blender [26]. By incorporating realistic underwater light, accurate depth maps, reference scene radiance, and unique underwater images, Underwater3k breaks away from the conventional underwater formation model. This allows us to effectively apply the synthetic dataset to real-world underwater environments and enhance the model's generalization capabilities. Moreover, experimental results on various real-world datasets exhibit the effectiveness and computational efficiency of our approach, with some examples shown in Fig. 1. We also evaluate the effectiveness of our LUIE methods on high-level tasks, i.e. underwater semantic segmentation and depth estimation. Additionally, a real-world test has conducted on a typical underwater ROV, proving the superior efficiency and effectiveness of LUIE in real-world settings.

The key contributions of this paper lie in three folds:

- We propose a learnable physical model-guided underwater image enhancement framework with bi-directional unsupervised domain adaptation, which can more accurately recover scene

radiance while reducing the domain gap between synthetic and real underwater scenes.

- We present a lightweight network architecture that is not only effective in reconstructing scene radiance but also highly efficient, ensuring rapid inference. The extensive experiments on public underwater datasets, real-world tests on underwater ROVs, and high-level tasks validate the superiority of our approach both in terms of quantitative and qualitative measures.
- We generate a new underwater dataset (Underwater3k) using Blender that accurately simulates underwater scenarios with realistic physical simulators. It contains underwater images, depth, background color maps, and referred clear underwater images.

The remaining sections of the paper are structured as follows. In Section 2, we provide a review of various methods employed for underwater image enhancement and restoration, as well as common domain adaptation techniques for enhancing underwater images. The proposed method is presented in Section 3. In Section 4, we present both qualitative and quantitative experimental results, including subjective and objective comparisons of different methods, demonstrations of downstream tasks utilizing our underwater image enhancement method, and ablation studies. Section 5 discusses the real-world test conducted. Finally, Section 6 offers the concluding remarks for this paper.

2. Related work

2.1. Underwater image enhancement

UIE methods can be broadly classified into two categories: prior-based [27–31] and learning-based methods [15,18,32–34]. Prior-based methods typically rely on physical models such as the retinex model [35] or atmospheric scattering model [24] or use techniques such as histogram equalization and color curve adjustment [27,28] to enhance underwater images. However, these methods are often challenging to generalize to complex underwater scenes and may produce enhanced images with artifacts or color casts.

In contrast, learning-based UIE methods have made significant progress, including supervised UIE [15,18,34,36], semi-supervised UIE [21], and unsupervised UIE [22,23], with the advent of DNNs. Li et al. [18] proposed a UIE method that focuses on enhancing underwater images in multiple color spaces, such as LAB, HSV, and RGB. This multi-color space approach allows for a more comprehensive enhancement of underwater images. However, many supervised UIE methods face the challenge of training their networks using a combination of synthetic and real underwater images. The gap between synthetic and real underwater images can make it difficult for the network to learn the accurate mapping of the real underwater enhancement process. To tackle this challenge, semi-supervised and unsupervised UIE methods have emerged. These methods aim to generalize the enhancement process to real underwater images by incorporating simplified physical

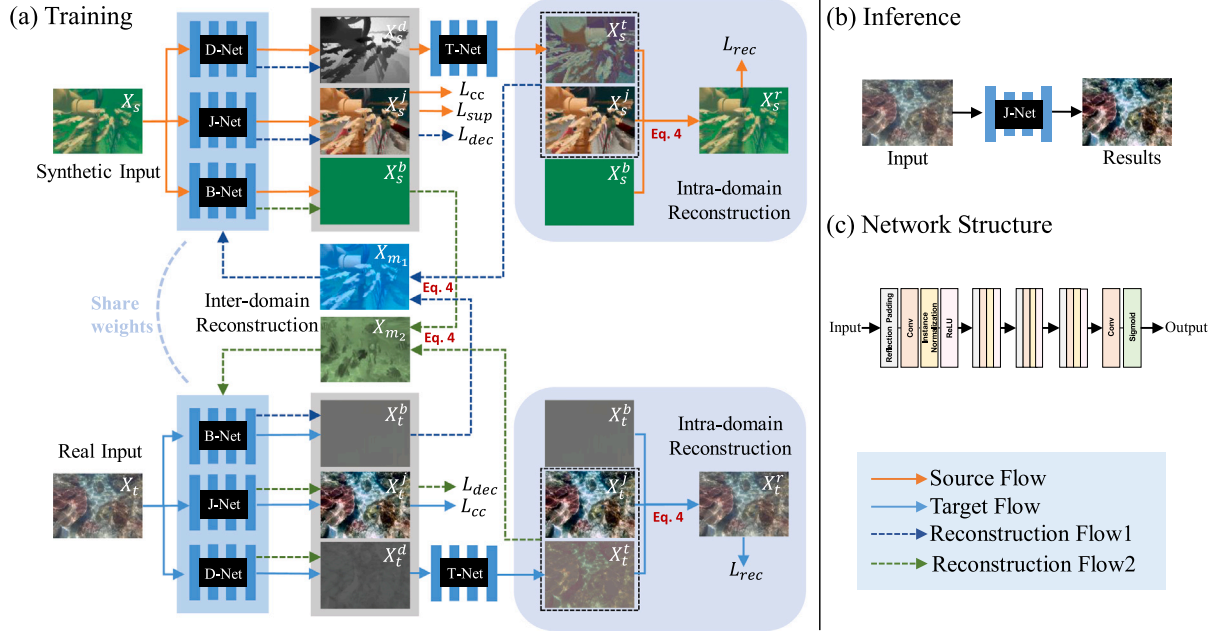


Fig. 2. The framework of our proposed LUIE. (a) illustrates the training process, with X_s and X_r representing labeled synthetic input (source) and unlabeled real input (target), respectively. X_s^d , X_s^l , and X_s^b represent the synthetic decomposed depth map, synthetic decomposed scene radiance, and synthetic decomposed background color, while X_r^d , X_r^l , and X_r^b represent their corresponding real decomposed components. X_s^l and X_r^l correspond to the synthetic and real transmission maps, while X_s^r and X_r^r represent the intra-domain reconstructed synthetic and real underwater images. X_{m1} and X_{m2} represent two inter-domain reconstructed underwater images. (b) shows the inference pipeline, where only the J-Net is utilized to predict the enhanced underwater images, ensuring computational efficiency. (c) presents the network structure of B-Net, J-Net, D-Net, and T-Net.

priors. By using simplified physical models, these methods attempt to obtain water-free images that can be used as references for enhancement. However, relying on simplified physical models can lead to inaccurate estimates of certain components, resulting in errors in the final enhanced underwater images. Therefore, we propose a novel unsupervised UIE method that incorporates a learnable physical model to construct more visual-friendly enhanced images without color cast and uneven exposure.

2.2. Domain adaptation for underwater image enhancement

Domain adaptation is a technique used to align the distribution shift between source and target domains. As for UIE tasks, several works have utilized domain adaptation to address the domain gap between synthetic and real underwater images. These approaches can be broadly categorized into three types. The first type involves using style transfer models, such as CycleGAN [37]. Jiang et al. [38] use the CycleGAN model to bridge the domain shift between synthetic and real underwater images, while Liu et al. [33] propose a recurrent translation framework to maintain consistency between synthetic and real underwater styles. The second type involves using adversarial learning to reduce the domain gap between different domains. Wen et al. [23] use both image translation and adversarial learning to transfer knowledge from the source domain to the target domain. The final type of domain adaptation method is contrastive learning [21], which learns invariant features between two different domains without introducing another network like the previous domain adaptation methods. However, we propose a novel bi-directional intra- and inter-domain adaptation method based on the underwater formation model. This approach does not require the introduction of another network, such as a discriminator, or relies on the transferring results.

3. Method

3.1. Framework overview

Our proposed method is based on unsupervised domain adaptation, with the aim of transferring knowledge from the labeled synthetic underwater domain to the unlabeled real underwater domain. As shown in Fig. 2, given a synthetic input X_s and a real input X_r , we utilize the B-Net, D-Net, and J-Net to decompose the underwater images into background color, depth map, and scene radiance. Our key focus is on obtaining accurate scene radiance, which represents clear underwater images without color cast or uneven exposure. To achieve this, we propose a learnable physical model that accurately models each underwater decomposed component. As the attenuation coefficients β are unattainable, we propose a learnable T-Net that directly learns the mapping between the depth map and the transmission map, which enables us to accurately obtain reconstructed underwater images using Eq. (1).

To further bridge the domain gap between synthetic and real underwater images, we propose a bi-directional intra- and inter-domain adaptation method inspired by image translation. Unlike existing methods that require additional networks, such as discriminators, we use simple recurrent reconstruction and decomposition to maintain consistency between the synthetic and real domains. Specifically, we reconstruct a new underwater image X_{m1} using the synthetic decomposed background color X_s^b , real transmission map X_r^l , and real scene radiance X_r^l . Similarly, we use X_r^b , X_s^l , and X_s^l to reconstruct another new underwater image X_{m2} based on our learnable physical model. These images are further decomposed into their original corresponding components, ensuring model consistency and reducing the domain gap between different domains.

In our method, the inference process is simplified by utilizing only the J-Net to obtain the recovered scene radiance, as depicted in Fig. 2(b). This eliminates the need for complex training settings or multiple networks, resulting in improved efficiency. This streamlined approach makes our method highly suitable for implementation on underwater ROVs.

3.2. Learnable underwater image formation model

We are familiar with the simplified atmospheric image formation model, which is represented by Eq. (1). In this model, the transmission map is given by:

$$t(x) = e^{-\beta d(x)}, \quad (2)$$

here, $d(x)$ represents the depth map, and β represents the attenuation coefficients that are difficult to obtain accurately in underwater scenes. Previous approaches have often simplified the transmission map in terms of β as a constant based on statistical data from experience, which is difficult to apply to all underwater scenes, which can lead to inaccurate estimates of scattered and absorbed light, resulting in further inaccuracies in predicted scene radiance, such as uneven exposure and color cast. Therefore, we use a network to learn the mapping of the depth map and transmission map as a learnable process. As shown in Fig. 2(a), we train the T-Net to obtain a more accurate transmission map as:

$$\begin{aligned} X_s^t &= f_t(X_s^d), \\ X_t^t &= f_t(X_t^d), \end{aligned} \quad (3)$$

where f_t is the learnable mapping of the depth map and transmission map by T-Net with the simple network structure in Fig. 2(b) as J-Net, which only contains five convolutional layers. Therefore we can rewrite the underwater image formation model as:

$$I(x) = J(x)f_t(d(x)) + A(x)(1 - f_t(d(x))). \quad (4)$$

Therefore, we introduce a novel learnable underwater image formation model that enhances the acquisition of precise transmission maps. This model enables accurate modeling of underwater scenes and facilitates the generation of transmission maps for various underwater environments. To ensure the accuracy of the depth map in real underwater scenarios, it is crucial to transfer knowledge from the source domain to the target domain. In this regard, we employ synthetic datasets called Underwater3K, which provide accurate depth maps for training the T-Net. This ensures that our model learns the depth information effectively.

Network structure. Our whole framework employs four networks for learning each component of the underwater image formation model as Eq. (4) to obtain the enhanced underwater images. This framework comprises B-Net, J-Net, D-Net, and T-Net, each adopting a similar network architecture to ensure efficient training and inference processes. As illustrated in Fig. 2(c), these networks are constructed with five convolutional layers and four instance normalization layers, with ReLU activation functions applied after each layer. Furthermore, A sigmoid function is used in the final layer to ensure the output range is between 0 and 1. This consistent architecture across networks is a deliberate choice to optimize for training and inference efficiency. It is important to note that during inference, only J-Net is utilized to generate clear underwater images, which substantially cuts down the inference time.

To further enhance the generalization capabilities of our model, we propose a bi-directional intra- and inter-domain adaptation approach, which allows our model to adapt to target domain, improving its ability to handle a wide range of underwater conditions.

3.3. Bi-directional intra- and inter-domain adaptation

To ensure the effectiveness of our learnable underwater image formation model and to reduce the domain gap between synthetic and real underwater branches, we introduce a bi-directional intra- and inter-domain adaptation method.

Intra-domain Decomposition and Reconstruction. Initially, we decompose the input into three components – background light, scene radiance, and depth map – using the B-Net, J-Net, and D-Net within an same efficient network structure, as illustrated in Fig. 2(b). The first component, background light, represents the corresponding color cast in underwater scenarios, while the second component, scene radiance, represents water-free images, and the depth map indicates the predicted depth of each object in the scene. To minimize the influence of color cast on the water, it is essential to ensure accurate modeling of underwater scenes. Consequently, we must guarantee that the decomposed components conform to the learnable underwater formation model, as expressed by Eq. (4). We reconstruct the synthetic and real branches to obtain reconstructed synthetic and real underwater images. These reconstructed underwater images are consistent with the corresponding input underwater images, ensuring the coherence of each network. By employing the $\mathcal{L}_{rec}$, we can not only maintain consistency between the input and reconstructed underwater images within the same domain but also effectively model the underwater scenes based on our learnable underwater formation model.

Inter-domain Reconstruction and Decomposition. The inter-domain adaptation method entails swapping the background light of each domain to reconstruct new underwater images using Eq. (5), which encourages us to perceive the background light as a domain-invariant content space, while the transmission and atmospheric light are treated as domain-specific attribute spaces. We can then recursively decompose the newly reconstructed images to obtain the same components as the original ones, utilizing the $\mathcal{L}_{dec}$. This process ensures that our networks learn domain-invariant features, thereby reducing the domain gap between synthetic and real domains, ultimately resulting in enhanced decomposition and reconstruction capabilities.

$$\begin{aligned} X_{m_1} &= X_s^j \cdot f_t(X_s^d) + X_t^b \cdot (1 - f_t(X_s^d)), \\ X_{m_2} &= X_t^j \cdot f_t(X_t^d) + X_s^b \cdot (1 - f_t(X_t^d)). \end{aligned} \quad (5)$$

3.4. Objective function

This section provides a description of the loss functions used in our proposed LUIE approach. We have categorized them into four types: synthetic image supervision loss, intra-domain reconstruction loss, inter-domain decomposition loss, and color constancy loss.

Synthetic Image Supervision Loss. As we model our synthetic underwater images in Blender, we are able to obtain accurate ground truth for these images. Therefore, we utilize the LPIPS [39] loss to guide our model using the synthetic ground truth.

$$\begin{aligned} \mathcal{L}_d &= \text{LPIPS}(X_s^d, Y_s^d), \\ \mathcal{L}_j &= \text{LPIPS}(X_s^j, Y_s^j), \\ \mathcal{L}_b &= \text{LPIPS}(X_s^b, Y_s^b), \end{aligned} \quad (6)$$

where $\mathcal{L}_d$ represents the supervised loss of the decomposed synthetic depth map X_s^d with synthetic depth ground truth Y_s^d , $\mathcal{L}_j$ represents the supervised loss of the decomposed synthetic scene radiance X_s^j with synthetic scene radiance ground truth Y_s^j , and $\mathcal{L}_b$ represents the supervised loss of the decomposed synthetic background color map X_s^b with synthetic background color ground truth Y_s^b . The overall supervised loss is given by:

$$\mathcal{L}_{sup} = \mathcal{L}_d + \mathcal{L}_j + \mathcal{L}_b. \quad (7)$$

Intra-domain Reconstruction Loss. In order to ensure that our decomposed components accurately follow the underwater image formation model, we strive for consistency between the reconstructed underwater

images obtained via Eq. (4) and the input underwater images using LPIPS loss.

$$\mathcal{L}_{rec} = \text{LPIPS}(X_s^r, X_s) + \text{LPIPS}(X_t^r, X_t). \quad (8)$$

Inter-domain Decomposition Loss. To further reduce the domain gap between synthetic and real underwater images, we introduce an inter-domain decomposition loss to learn the domain-invariant features.

$$\begin{aligned} \mathcal{L}_{dec} = & \text{LPIPS}(X_s^{td}, X_s^d) + \text{LPIPS}(X_t^{td}, X_t^d) \\ & + \text{LPIPS}(X_s^{tj}, X_s^j) + \text{LPIPS}(X_t^{tj}, X_t^j) \\ & + \text{LPIPS}(X_s^{tb}, X_s^b) + \text{LPIPS}(X_t^{tb}, X_t^b), \end{aligned} \quad (9)$$

here, X_s^{tb} , X_s^{tj} , and X_s^{td} represent the decomposed components of X_{m_1} , while X_t^{tb} , X_t^{tj} , and X_t^{td} represent the decomposed components of X_{m_2} .

Color Constancy Loss. To further reduce the color cast in our decomposed scene radiance, we introduce the color constancy loss, which is based on the Gray World assumption [22,40,41]. This assumption promotes a balance among the intensity values of the three color channels in the scene radiance.

$$\mathcal{L}_{cc} = \sum_{c_1, c_2} (|J_{c_1} - J_{c_2}|_1), \quad (10)$$

where (c_1, c_2) represents the combination of two color channels in RGB, and J encompasses both X_s^j and X_t^j .

In conclusion, we can express our overall optimization objective as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{sup} + \alpha \cdot \mathcal{L}_{rec} + \beta \cdot \mathcal{L}_{dec} + \gamma \cdot \mathcal{L}_{cc}, \quad (11)$$

where α , β , and γ serve as the hyperparameters for loss weight. In this work, we set the values as follows: $\alpha = 1.0$, $\beta = 1.0$, and $\gamma = 0.1$.

4. Experiment results

4.1. Datasets

We evaluate the performance of our method on four widely used public underwater image enhancement datasets: UIEB [15], LSUI [34], RUIE [42], and C60 [15]. Additionally, we apply our method to real-world underwater scenes that we have collected for real-world tests. **UIEB dataset [15].** The UIEB dataset consists of 890 real-world underwater images, along with their corresponding ground truth images. It is worth mentioning that the reference images in UIEB are subjective and obtained through existing methods or commercial applications. To create a training and testing split, we partition the UIEB dataset into 700 training images (excluding the use of ground truth) and 190 testing images.

RUIE dataset [42]. RUIE is a large-scale real-world underwater image dataset that includes three subsets of quality images. However, these subsets do not come with ground truth images.

C60 dataset [15]. C60 is a challenging underwater dataset derived from UIEB and contains no-reference images. This dataset poses additional difficulties in underwater image enhancement tasks.

LSUI dataset [34]. LSUI is a comprehensive large-scale underwater dataset that includes a diverse range of images sourced from various methods. These methods include public datasets, websites, and real-world collections, which provides a wide variety of underwater scenes, enabling robust evaluation and training of underwater image enhancement methods.

Underwater3k. We utilize Blender to create the Underwater3k dataset, consisting of 7 scenes, each with 4 distinct background colors and randomly assigned attenuation coefficients. By leveraging Blender's scene construction capabilities, we were able to generate accurate depth maps and background colors. Several examples of our synthetic data are presented in Fig. 3.

For training our method, we utilize 2800 labeled images from our synthetic underwater image pairs. Additionally, we include 2800 unlabeled images from UIEB, LSUI, and RUIE datasets. This combination of labeled and unlabeled data aids in improving the performance and generalization of our method.

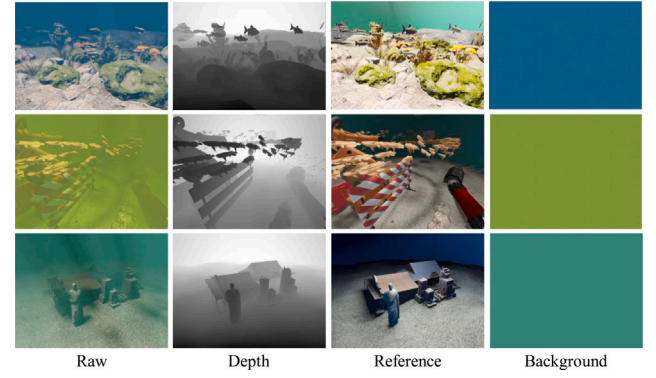


Fig. 3. The examples of synthetic datasets Underwater3k.

4.2. Implementation details

The whole LUIE framework is implemented on Pytorch with a single NVIDIA RTX A6000 GPU. We employ the ADAM [45] optimizer to optimize our LUIE with the learning rate of 2×10^{-4} , the batch size is 1, and the epoch is 200. For our network structure, the channel size of each convolutional layer is 64 while kernel size is 3×3 . During the training optimization, we employ the LPIPS with layers from the pre-trained AlexNet [46].

4.3. Comparison with other underwater image enhancement methods

We compare our proposed LUIE method with the following underwater image enhancement methods: (1) prior-based UIE methods: GDCP [30]; (2) supervised UIE methods: WaterNet [15], GL-Net [43], UWCNN [25], FUnIE-GAN [20], Ucolor [18], MetaUE (fine-tuning on UIEB) [19]; (3) semi-supervised UIE methods: Semi-UIR [21]; (4) unsupervised methods: UIE-DAL [44], USUIR [22], SyreaNet [23]. We use the official codes and checkpoints provided by the authors for fair comparisons. It is important to note that the methods UIE-DAL [44] and UWCNN [15] necessitate the designation of specific water types for the datasets they are applied to. Therefore, for the purpose of evaluating the public datasets [15,42], we have chosen the median water type, labeled as “type 5”, to align with the datasets’ conditions.

4.3.1. Quantitative comparisons

We evaluate the performance of our proposed LUIE method, along with several other existing state-of-the-art methods, on three benchmark underwater image enhancement datasets: UIEB, RUIE, and C60. We employ both full-reference evaluation metrics, including PSNR, SSIM [47], and LPIPS [39], as well as no-reference evaluation metrics that focus on underwater image quality assessment (UIQA), such as UIQM [48], UCIQE [49], and Uranker [50]. Table 1 presents the quantitative results of our LUIE method compared to the other methods on these datasets.

Full-reference evaluation results. In terms of full-reference evaluation, our proposed LUIE method significantly outperforms unsupervised methods and demonstrates a slight advantage over supervised and semi-supervised methods across all the metrics. These results indicate that LUIE achieves superior performance in accurately enhancing underwater images compared to other approaches. Furthermore, we provide two supplementary metrics, DeltaE and HDR-VDP-2 [51], which gauge the perceptual difference between two colors and assess the visibility and image quality of two images in Table 2. These metrics serve as further evidence supporting the effectiveness of our LUIE model in accurately recovering scene radiance in full-reference datasets.

No-reference evaluation results. Regarding the no-reference evaluation results, LUIE achieves competitive performance across various

Table 1

The quantitative comparisons on UIEB [15], RUIE [42], and C60 [15] dataset. ‘PB’ represents the prior-based method, ‘SL’ represents the supervised method, ‘SSL’ represents the semi-supervised method, and ‘UL’ represents the unsupervised method. The best score is in **bold**, and the second score is in underline.

Method	Types	UIEB [15]				RUIE [42]			C60 [15]		
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Uranger $\uparrow$	UIQM $\uparrow$	UCIQM $\uparrow$	Uranger $\uparrow$	UIQM $\uparrow$	UCIQM $\uparrow$	Uranger $\uparrow$
Input	–	17.18	0.7514	0.2682	1.736	4.025	28.82	1.433	3.108	25.38	–0.1372
GDCP [30]	PB	<u>13.03</u>	<u>0.7446</u>	<u>0.3361</u>	<u>3.652</u>	<u>3.652</u>	<u>30.17</u>	<u>1.264</u>	<u>4.240</u>	<u>29.75</u>	<u>0.4471</u>
WaterNet [15]		22.08	0.8768	0.1329	2.410	5.218	30.45	2.241	4.256	30.61	1.202
GL-Net [43]		22.29	0.9037	<u>0.1312</u>	2.912	5.170	31.25	2.600	4.149	30.21	1.623
UWCNN [25]	SL	17.74	0.7470	0.3340	1.539	5.144	22.96	1.549	4.206	23.15	0.5120
FUnIE-GAN [20]		17.78	0.7604	0.3170	1.743	3.415	30.74	1.935	3.061	28.58	0.4242
Ucolor [18]		<u>22.46</u>	0.8929	0.1364	2.577	5.126	29.99	2.239	4.024	28.34	0.9667
MetaUE [19]		22.11	0.9045	0.1836	2.756	5.188	30.52	2.864	3.883	30.45	1.728
Semi-UIR [21]	SSL	<u>22.25</u>	<u>0.9090</u>	<u>0.1581</u>	<u>2.900</u>	<u>4.953</u>	<u>30.34</u>	<u>2.703</u>	<u>3.621</u>	<u>30.53</u>	<u>1.538</u>
UIE-DAL [44]		18.37	0.8492	0.1901	1.921	4.778	29.97	2.330	4.215	29.18	0.8894
USUIR [22]	UL	21.73	0.8820	0.1763	2.392	4.777	31.53	2.184	4.082	<u>30.56</u>	0.9518
SyreaNet [23]		18.04	0.8629	0.1996	2.572	5.724	29.68	2.485	4.870	28.51	1.340
LUIE (ours)	UL	22.87	0.9167	0.1241	2.906	<u>5.246</u>	<u>31.33</u>	2.590	<u>4.358</u>	30.62	<u>1.625</u>

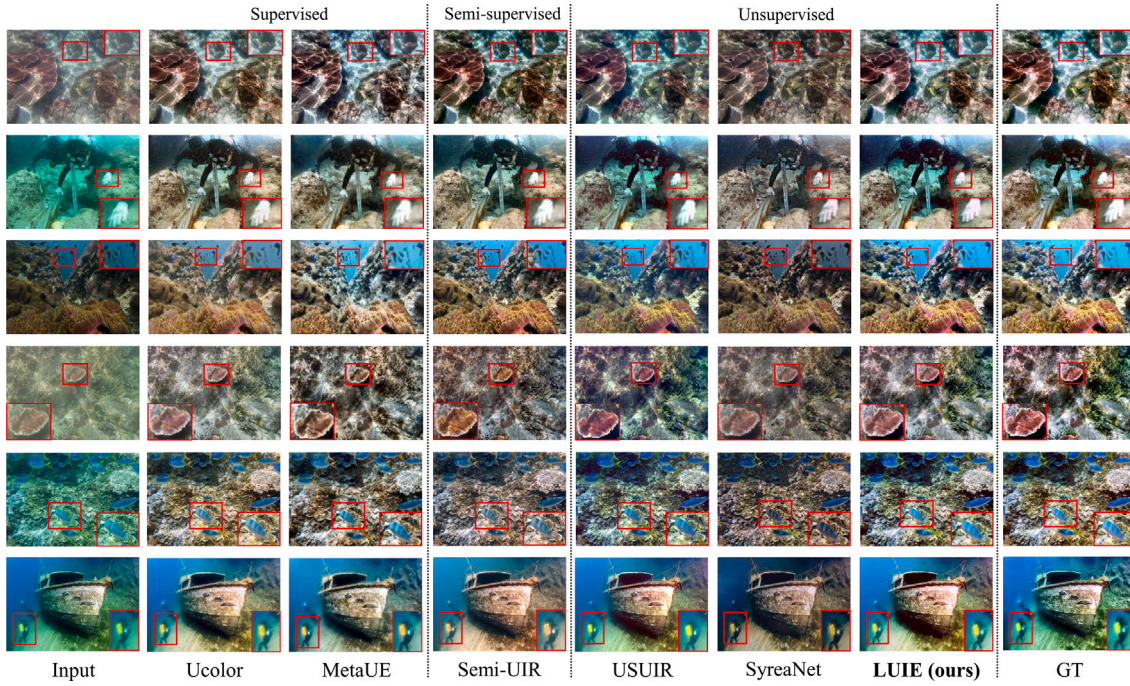


Fig. 4. The qualitative comparisons of full-reference data on UIEB [15] dataset. **Better viewed when zoomed in.**

Table 2

The additional quantitative results on UIEB [15] datasets.

Method	WaterNet	Ucolor	Semi-UIR	USUIR	SyreaNet	LUIE (ours)
DeltaE $\downarrow$	9.064	8.968	8.425	8.367	15.05	7.957
HDR-VDP-2 $\uparrow$	92.21	92.28	92.44	92.41	91.52	92.62

UIQA metrics. This demonstrates its ability to effectively assess and enhance the quality of underwater images without relying on reference images. In the RUIE [42] datasets, our LUIE model achieves the second highest scores in the UIQM and UCIQM metrics. Furthermore, in the C60 [15] datasets, our LUIE model attains the best score in the UCIQM metric and the second highest scores in the UIQM and Uranger metrics. To further evaluate the quality of the improved underwater images produced by various methods, especially in this competitive environment, we have introduced a general no-reference image quality assessment (IQA) metric that has been trained with a large dataset of natural images, namely CNNIQA [52], as shown in Table 3. This result indicates that our methods achieve better image quality with general IQA. It is worth noting that these no-reference UIQA metrics are primarily

Table 3

The additional quantitative results on C60 [15] and RUIE [42] datasets with CNNIQA [52].

Method	MetaUE [19]	USUIR [22]	SyreaNet [23]	LUIE (ours)
C60	5.207	5.127	<u>6.103</u>	6.617
RUIE	<u>6.379</u>	5.815	5.161	6.797

subjective. Therefore, we provide an objective evaluation of our LUIE model’s ability to enhance underwater images for assisting high-level tasks in Section 4.4, further demonstrating its effectiveness. Overall, LUIE demonstrates better performance as an unsupervised method, surpassing not only supervised approaches but also semi-supervised methods.

4.3.2. Qualitative comparisons on full-reference dataset

We evaluate the visual results of different methods in Fig. 4. Both USUIR and SyreaNet exhibit noticeable color cast issues in their results, failing to accurately correct distorted colors, especially under conditions of poor illumination, as evidenced in the images from the

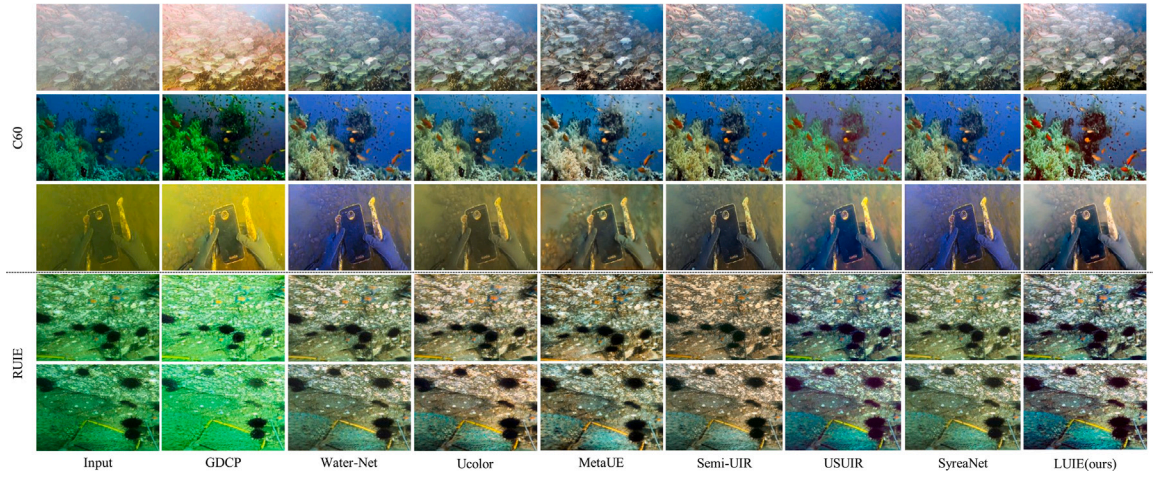


Fig. 5. The qualitative comparisons of no-reference data on C60 [15] and RUIE [42]. Better viewed when zoomed in.

Table 4

The computational efficiency comparisons on UIEB test sets with the resolution of 1024×1024 .

Methods	Model parameters (M)	Inference time (s)
WaterNet	24.81	0.0771
FUnIE-GAN	7.02	0.0063
Ucolor	157.40	0.1779
MetaUE	24.42	0.0140
Semi-UIR	1.68	0.0332
USUIR	0.23	0.0055
SyreaNet	29.06	0.0487
LUIE (ours)	0.11	0.0024

third row. While other supervised or semi-supervised techniques may produce colors closer to the ground truth, they too grapple with issues such as low contrast or inconsistent exposure. Moreover, these methods may not be robust enough to cope with the full types of underwater imaging environments. For example, they often fall short in restoring the correct colors in scenarios with an extremely greenish tint, as seen in the fourth and fifth rows. In comparison, our proposed LUIE method attains accurate color and uniform exposure results, which can be attributed to its more precise modeling of underwater scenes, distinguishing it from other methods.

4.3.3. Qualitative comparisons on no-reference datasets

The visual results of our proposed method on the no-reference datasets RUIE and C60 can be seen in Fig. 5. The visual outcomes produced by GDCP [30] suffer from issues of uneven exposure. Supervised and semi-supervised methods, such as WaterNet, Ucolor, MetaUE, and Semi-UIR, struggle with generalization in complex underwater scenarios that exhibit both high color saturation and insufficient illumination, as demonstrated in the second and third rows. Additionally, unsupervised methods tend to result in enhanced images with uneven contrast across various regions. Our method effectively restores the exposure of the images and achieves satisfying color correction. In particular, in the third row of C60, our enhanced image significantly reduces the impact of the yellow color cast and uneven exposure issues, resulting in a more visually appealing image. The experimental results demonstrate the robustness of our methods across various complex underwater scenarios.

4.3.4. Comparisons on computational efficiency

Experiments were conducted to evaluate the computational efficiency of our proposed method and other existing underwater image enhancement methods. We use the UIEB test set, comprising 190

Table 5

The quantitative results of semantic segmentation.

Method	Accuracy	F1	mIoU
SUIMNet [1]	0.7880	0.7328	0.7565
SUIMNet + MetaUE [19]	0.8278	0.8892	0.7865
SUIMNet + SyreaNet [23]	0.7281	0.7282	0.7276
SUIMNet + ours	0.9215	0.9108	0.8653

images with a resolution of $1,024 \times 1,024$, and measure the inference time on a single NVIDIA RTX A6000 GPU. Table 4 presents the results, showing that our proposed method exhibited the smallest number of parameters and the fastest inference time among all the evaluated methods. Despite its computational efficiency, our proposed method also achieved the best results in terms of image enhancement, as illustrated in Fig. 1(c). These experiments highlight the superior computational efficiency of our proposed method. It not only requires fewer parameters and achieves faster inference times but also delivers outstanding performance in enhancing underwater images. These characteristics make our method suitable for real-time applications where computational efficiency is crucial.

4.4. High-level task performance

We conduct additional experiments to quantitatively evaluate the effectiveness of our proposed LUIE for high-level tasks.

Semantic segmentation. We use the SUIM [1] underwater segmentation dataset and compare the accuracy, F1 score, and mIoU of segmentation results generated by SUIMNet [1] using raw underwater images and SUIMNet [1] using enhanced underwater images from supervised method MetaUE [19], unsupervised method SyreaNet [23] and our proposed LUIE, as shown in Table 5. The quantitative results show the effectiveness of the SUIMNet with our LUIE-enhanced underwater images compared with other state-of-the-art methods. Additionally, Fig. 6 illustrates the SUIMNet, when applied to our LUIE, produces more detailed and accurate segmentation predictions. The significant improvement in underwater semantic segmentation results, as measured by each quantitative metric, confirms the effectiveness of our LUIE method.

Depth Estimation. Underwater depth estimation is a crucial component for various underwater applications, including underwater semantic segmentation and the effective inspection of ROVs. In our LUIE approach, we incorporate the D-Net, a depth estimation network, which we evaluate using the USOD10K dataset [53]. Additionally, the enhanced underwater images produced by our LUIE model can also assist other depth estimation networks, such as UDepth [54].

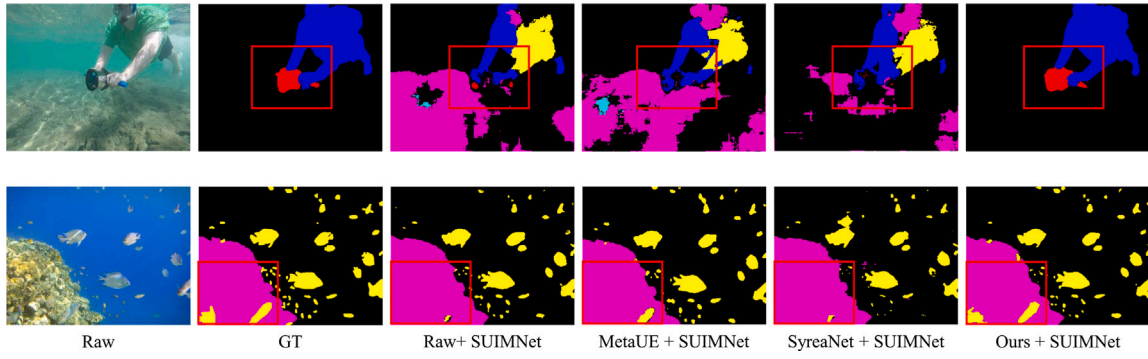


Fig. 6. The semantic segmentation results of SUIM [1] datasets.

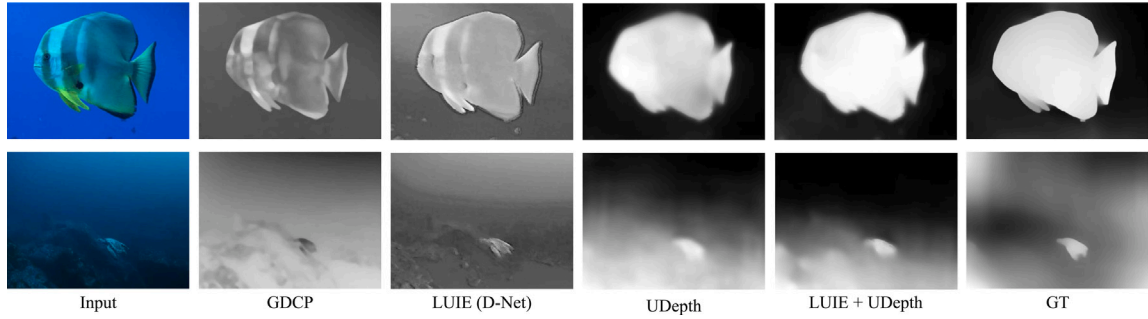


Fig. 7. The depth estimation results of USOD10K [53] datasets.

Table 6
The quantitative results of depth estimation.

Method		RMSE ↓	EPE ↓
Direct	GDCP [30]	0.6401	7.113
	Ours (D-Net)	0.6201	6.901
Assist	UDepth [54]	0.5333	5.974
	UDepth + ours	0.4917	5.517

To evaluate the accuracy and performance of our depth estimation module and the assisting ability of our LUIE model, we compared the results using Root mean squared error (RMSE) and End Point Error (EPE) metrics [55]. Table 6 and Fig. 7 present a comprehensive comparison between different approaches for generating depth maps. The evaluation includes directly using the enhancement method to generate depth maps, specifically our D-Net and GDCP, as well as utilizing a depth estimation network, such as UDepth, with both raw underwater images and our enhanced underwater images. The results clearly demonstrate that our LUIE method achieves significantly more accurate depth estimations, whether in directly using D-Net or in assisting depth estimation network, closely aligning with the ground truth. This improvement in depth estimation demonstrates the effectiveness of our LUIE method in enhancing the overall performance of underwater tasks. By accurately estimating underwater depth, our method provides valuable information for applications such as underwater semantic segmentation and ROV inspections, enabling more precise and reliable underwater analysis and decision-making.

Discussion. Based on the aforementioned experiments, we observe that our LUIE model possesses the ability to assist high-level tasks. This indicates the potential for extending our enhancement network to simultaneously enhance underwater images while performing tasks such as semantic segmentation, object detection, and depth estimation, as shown in Fig. 8. By incorporating high-level task constraints into the enhancement network and allowing them to mutually benefit and improve each other, we can explore promising avenues for future work. This integration and synergy between the enhancement network and

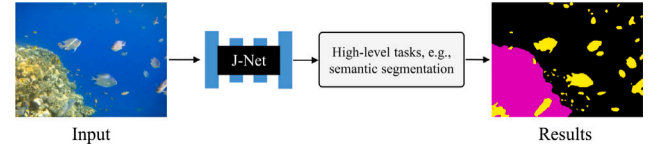


Fig. 8. Multi-task framework for underwater image enhancement and high-level tasks.

high-level tasks hold significant potential for advancing the field of underwater image enhancement.

4.5. Ablation study

4.5.1. The effectiveness of each component and loss

To determine the effectiveness of each component and loss, we conduct an ablation study on the UIEB test set, evaluating PSNR, SSIM, and LPIPS. In Table 7, a base model is established using only $\mathcal{L}_{sup}$, without any domain adaptation strategy or physical model, and the corresponding numerical scores are presented in the first row. Excluding $\mathcal{L}_{rec}$, which involves intra-domain reconstruction, results in a PSNR of 20.38. Excluding $\mathcal{L}_{dec}$ yields a PSNR of 21.81. However, by excluding the color constancy loss, $\mathcal{L}_{cc}$, a PSNR of 22.61 is achieved, with improvements in each metric. Ultimately, all framework components are added to obtain the final results and achieve the best performance. Furthermore, we provide a visualization of the ablation study for each component in Fig. 9, allowing for a direct observation of the individual effects of these components.

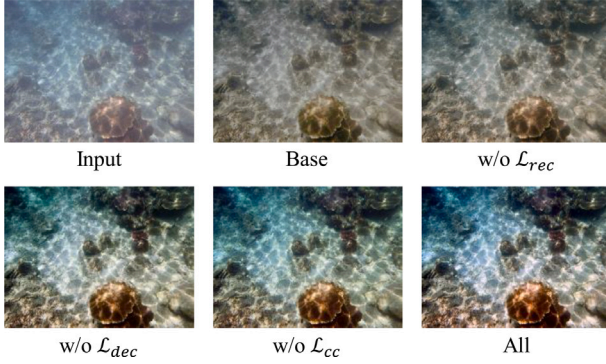
4.5.2. The effectiveness of the learnable physical model

To assess the effectiveness of the learnable physical model, different methods are employed to obtain reconstructed underwater images. Initially, the simplified physical model (Eq. (1)) is used to reconstruct the decomposed components of underwater images, resulting in a PSNR of 19.38, as shown in the first row of Table 8. Subsequently, a multi-to-one network is developed as a black-box model to reconstruct the

Table 7

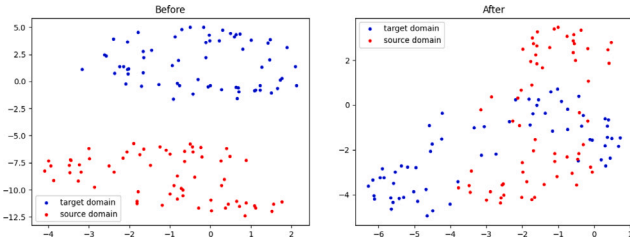
The ablation study for components and losses.

Ablations	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Base	17.46	0.7170	0.2854
w/o $\mathcal{L}_{rec}$	20.38	0.8976	0.1843
w/o $\mathcal{L}_{dec}$	21.81	0.9049	0.1426
w/o $\mathcal{L}_{cc}$	22.61	0.9132	0.1331
All	22.87	0.9167	0.1241

**Fig. 9.** The visualization of ablation study for components and losses in our proposed LUIE.**Table 8**

The ablation study for the learnable physical model.

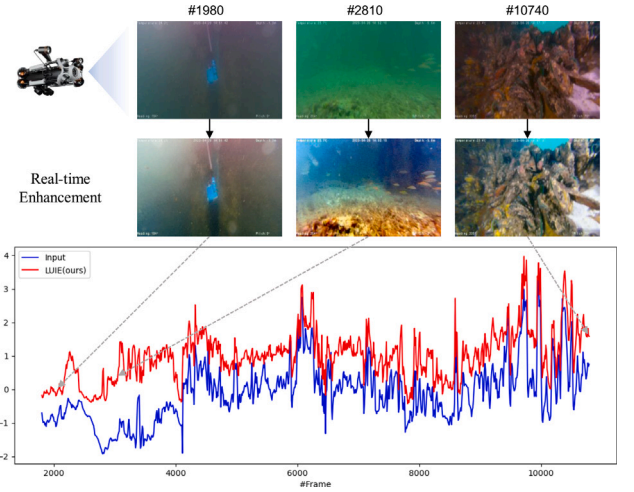
Methods	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Simplified physical model	19.38	0.8574	0.2177
Black-box model	20.72	0.8610	0.1488
Learnable physical model	22.87	0.9167	0.1241

**Fig. 10.** The t-SNE [56] plots compare the data distribution between the source domain data and the target domain data. We analyze the distribution before and after applying our domain adaptation method with LUIE.

three decomposed components into an underwater image. Although this model exhibits a slight improvement over the simplified physical model, it still performs worse than the learnable physical model. These experiments clearly demonstrate the effectiveness and importance of our learnable physical model.

4.5.3. Visualization of data distribution

Fig. 10 presents t-SNE [56] plots that illustrate the distribution of our data between the source domain (synthetic data) and the target domain (real underwater data). In the original state, we observe a significant domain shift between the two domains, with a large distance between their respective feature distributions. However, after applying our LUIE method, we can observe that the distances between the features become much closer, indicating a substantial reduction in the domain gap between the source and target domains. This demonstrates the effectiveness of our LUIE approach in bridging the gap between synthetic and real underwater data distributions.

**Fig. 11.** Real-world underwater ROV inspection tests are carried out at various depths in underwater environments. The red and blue lines represent the Unranker results for our enhanced images and the input original images, respectively. The experimental results confirm the robust performance of LUIE in these scenarios.

5. Real-world tests

Real-World Test Setup. To validate the effectiveness and efficiency of our LUIE model in real-world underwater environments, we conduct experiments using the CHASING M2 PRO MAX underwater ROV [57], equipped with the NVIDIA Jetson AGX Xavier (32 GB GPU and NVIDIA Carmel ARMv8.2 CPU) platform.

We implement the J-Net component of our proposed LUIE model on the NVIDIA Jetson AGX Xavier platform to ensure real-time performance. The LUIE model achieve an impressive processing rate of 132.84 frames per second (FPS) on the NVIDIA Jetson AGX Xavier (32 GB) GPU. Additionally, we test the LUIE model on an NVIDIA Carmel ARMv8.2 CPU, where it processed frames at a rate of 2.88 FPS. These computational capabilities make our LUIE model suitable for real-time underwater ROV inspections, providing efficient and responsive image enhancement. In our real-world tests, we evaluate the LUIE model at various underwater depths, considering diverse color casts and illuminations. Fig. 11 showcases some examples of the results obtained from our real-world tests. These tests confirm the feasibility and robustness of our proposed LUIE method in enhancing underwater images captured in challenging real-world environments. The successful implementation of our LUIE model on the CHASING M2 PRO MAX underwater ROV, along with its impressive processing speed and consistent performance across different underwater conditions, demonstrates the practical applicability and reliability of our method for real-time underwater image enhancement tasks.

Future Work. The network designed for real-time underwater image enhancement can be integrated with more advanced and functional applications geared toward underwater exploration. These applications include but are not limited to, underwater object detection and underwater Simultaneous Localization and Mapping (SLAM). Furthermore, to guarantee the robust performance and practical utility of these enhancement methods within real-world scenarios, it is essential to account for the challenges presented by complex underwater environments, including low-light conditions and complex color casts.

6. Conclusion

We proposed a comprehensive unsupervised framework to enhance real-world underwater images, which includes a novel learnable physical model-guided UIE with bi-directional domain adaptation, dubbed LUIE. Our learnable physical model promotes better underwater color

accuracy and more even exposure. Moreover, the bi-directional domain adaptation technique effectively minimizes the domain gap between synthetic and real underwater datasets, thus improving the generalization ability of our method. We also created a new underwater dataset, Underwater3k, using Blender to generate underwater scene images without converting indoor datasets to underwater images. Notably, our approach offers a solution with lower computational costs and greater generalization abilities during inference, making it well-suited for practical underwater ROV applications. High-level tasks, *i.e.*, semantic segmentation and depth estimation, also validated the effectiveness of our method. Additionally, we further validated the real-time and robust capabilities of our proposed methods with real-world underwater ROV settings.

CRedit authorship contribution statement

Jingyi Pan: Writing – original draft, Software, Methodology. **Zeyu Duan:** Writing – review & editing, Resources. **Jianghua Duan:** Writing – review & editing, Supervision. **Zhe Wang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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